

3 Dynamics, Stability, and Basic Lyapunov Theory

3.1 Dynamical Systems

We will (almost) exclusively work with discrete-time dynamical systems described by a difference equation

$$x_{t+1} = f(x_t, u_t), \quad x_0 = x_{\text{init}}, \quad t = 0, 1, 2, \dots, \quad (3.1)$$

which we will also write as

$$x_+ = f(x, u). \quad (3.2)$$

Here $x \in \mathbb{R}^n$ is the state, $u \in \mathbb{R}^m$ is the control input, and x_+ is shorthand for the value of the state at the next time step.

Disturbances and time-dependence A more general version of (3.2) would allow for time-dependence and disturbances,

$$x_{t+1} = f_t(x_t, u_t, w_t), \quad u_t = \pi_t(x_t), \quad (3.3)$$

where w_t is a disturbance and both the dynamics f_t and the policy π_t are allowed to vary with time. While we will sometimes need disturbances and time-dependence, the vast majority of our analysis will be focused on the time-invariant, disturbance-free form (3.2).

3.2 Open Loop and Closed Loop Systems

We will mostly study systems in which the control input is generated by a *state feedback policy* π , together with an auxiliary input v , i.e. $u = \pi(x) + v$. Under this structure, the system (3.2) is said to be in *open loop* if $\pi(x) \equiv 0$, giving $x_+ = f(x, v)$. If we set $v \equiv 0$ as well, this gives the *unforced* open loop system

$$x_+ = f(x, 0) =: f(x). \quad (3.4)$$

Such a system, depending solely on the state and no control inputs, is also called an *autonomous system*.¹ Applying a policy $u = \pi(x) + v$ gives the *closed loop* system

$$x_+ = f(x, \pi(x) + v) =: f^\pi(x, v). \quad (3.5)$$

When $u = \pi(x)$, i.e., $v \equiv 0$, we write $x_+ = f^\pi(x, 0) =: f^\pi(x)$.²

¹An autonomous system can equivalently be defined simply as $x_+ = f(x)$, without the need for any control inputs.

²This is the same closed loop notation introduced in the *Two Motivating Problems* topic.

3.3 A First Question

We begin our discussion by focusing on settings where the dynamics depend exclusively on x , i.e., $u \equiv 0$ or $u = \pi(x)$. We will mostly be interested in *initial value problems*, where x_t describes how $x_{t+1} = f(x_t)$ evolves over $t = 0, 1, 2, \dots$ from a known initial condition $x_0 = x_{\text{init}}$. A first question we might ask is:

Key Question 3.1. Suppose that we want to learn something from data generated by a dynamical system $x_+ = f(x)$. What properties of the system $x_+ = f(x)$ might make this easier?

This is the same question, posed more generally, that opened both of our motivating problems: what makes $\hat{f} \approx f$ (or $\hat{\pi} \approx \pi^*$) translate into trajectories that stay close to the true ones? The rest of this chapter develops the vocabulary — stability — needed to start answering it.

3.4 Stability

The intuition we are after: generalization is only possible if nearby samples are similar, i.e., if a small change to a system's input leads to a correspondingly small change to its output. When we cover elements of statistical learning theory, we will see that this is often captured by a Lipschitz constant. In the context of the problems we are studying, the “input” to a system is the initial condition, and the output is the resulting trajectory. Therefore we need to capture a notion of smoothness at the trajectory level. We begin with the simplest version of this idea, where all trajectories of a system (locally) converge to the same point.

Definition 3.1 (Equilibrium point). A point x_{eq} is an *equilibrium point* of the system $x_+ = f(x)$ if $f(x_{eq}) = x_{eq}$, i.e., if it is a fixed point of the map f .

We will assume, without loss of generality, that 0 is an equilibrium point of $x_+ = f(x)$, i.e., $f(0) = 0$.

Definition 3.2 (Stability, cf. Khalil [Kha02, Def. 4.1]). The equilibrium point $x = 0$ of $x_+ = f(x)$ is

- *stable* if, for each $\varepsilon > 0$, there is $\delta = \delta(\varepsilon) > 0$ such that

$$\|x_0\| < \delta \implies \|x_t\| < \varepsilon, \quad \forall t \geq 0; \quad (3.6)$$

- *unstable* if it is not stable;
- *asymptotically stable* if it is stable and δ can be chosen such that

$$\|x_0\| < \delta \implies \lim_{t \rightarrow \infty} x_t = 0. \quad (3.7)$$

Remark 3.3. As stated, stability and asymptotic stability are *local* notions: they only need to hold for x_0 within some (possibly small) δ -ball around the equilibrium. We write LAS for asymptotic stability in this sense, to emphasize the possibly-local domain on which it holds. When δ can be taken arbitrarily large – equivalently, when the property holds for every $x_0 \in \mathbb{R}^n$ – we say the origin is *globally stable*, or *globally asymptotically stable* (GAS).

Definition 3.2 is precise, but not directly useful for computation: verifying it requires reasoning about every ε and producing a matching δ . The following comparison functions let us restate it in a form we can actually compute with.

Definition 3.4 (Class $\mathcal{K}$ and $\mathcal{K}_\infty$ functions). A continuous function $\alpha : [0, a) \rightarrow [0, \infty)$ is of class $\mathcal{K}$ if it is strictly increasing and $\alpha(0) = 0$. It is of class $\mathcal{K}_\infty$ if, in addition, $a = \infty$ and $\alpha(r) \rightarrow \infty$ as $r \rightarrow \infty$.

Definition 3.5 (Class $\mathcal{KL}$ function). A function $\beta : [0, \infty) \times [0, \infty) \rightarrow [0, \infty)$ is of class $\mathcal{KL}$ if, for each fixed $s \geq 0$, $\beta(\cdot, s)$ is of class $\mathcal{K}$, and for each fixed $r \geq 0$, $\beta(r, \cdot)$ is decreasing in s with $\beta(r, s) \rightarrow 0$ as $s \rightarrow \infty$.

Lemma 3.6 (Class $\mathcal{K}/\mathcal{KL}$ characterization of stability, cf. Khalil [Kha02, Lemma 4.5], specialized to autonomous systems). *The equilibrium point $x = 0$ of $x_+ = f(x)$ is*

- *stable if and only if there exist a class $\mathcal{K}$ function α and a constant $c > 0$ such that*

$$\|x_t\| \leq \alpha(\|x_0\|), \quad \forall t \geq 0, \forall \|x_0\| < c; \quad (3.8)$$

- *asymptotically stable if and only if there exist a class $\mathcal{KL}$ function β and a constant $c > 0$ such that*

$$\|x_t\| \leq \beta(\|x_0\|, t), \quad \forall t \geq 0, \forall \|x_0\| < c; \quad (3.9)$$

- *globally asymptotically stable if and only if the second inequality holds for any initial state x_0 .*

Remark 3.7. This is Khalil's Lemma 4.5, restricted to time-invariant (autonomous) systems.

Specializing the class $\mathcal{KL}$ characterization of asymptotic stability above to $\beta(r, s) = C\rho^s r$, for some $C > 0$, $\rho \in [0, 1)$, gives a stronger, and often more useful, notion:

Definition 3.8 (Exponential stability). The equilibrium $x = 0$ of $x_+ = f(x)$ is *exponentially stable* if there exist $C > 0$, $\rho \in [0, 1)$ such that

$$\|x_t\| \leq \beta(\|x_0\|, t) = C\rho^t \|x_0\| \quad (3.10)$$

for all $\|x_0\| \leq B_0$. It is *globally exponentially stable* (GES) if this holds for any $B_0 > 0$.

A linear system $x_+ = Ax$ is GAS if and only if it is LAS if and only if it is GES if and only if $\max_i |\lambda_i(A)| < 1$.

💡 *Key Result 3.1*

Remark 3.9. For linear systems, every notion of stability we define collapses to the single spectral condition $\max_i |\lambda_i(A)| < 1$, and is automatically *global* and easy to verify. None of this survives once we leave the linear world — local and global stability genuinely differ, and verifying either becomes the hard part.

3.5 Basic Lyapunov Theory

Lyapunov theory provides a general approach to certifying various notions of stability of a nonlinear system. A comprehensive treatment of Lyapunov theory could easily take up an entire semester: instead, we aim here to briefly convey some key concepts and ideas. While we will rarely directly use these tools, they are nevertheless crucial in certifying the various stability assumptions we will need in the sequel.

Theorem 3.10 (Lyapunov's direct method, discrete time). *Let $x = 0$ be an equilibrium point for the system $x_+ = f(x)$,^a with $f : D \rightarrow \mathbb{R}^n$, $0 \in D \subset \mathbb{R}^n$, locally Lipschitz in x on D . Let $V : D \rightarrow \mathbb{R}_+$ be a continuous function such that*

$$\alpha_1(\|x\|) \leq V(x) \leq \alpha_2(\|x\|), \quad (3.11)$$

$$V(f(x)) - V(x) \leq -\alpha_3(\|x\|), \quad (3.12)$$

for all $x \in D$, where $\alpha_1, \alpha_2, \alpha_3$ are class $\mathcal{K}$ functions. Then $x_+ = f(x)$ is LAS, and there exists $\beta \in \mathcal{KL}$ such that

$$\|x_t\| \leq \beta(\|x_0\|, t) \quad \text{for all } t = 0, 1, \dots \quad (3.13)$$

Furthermore, if $D = \mathbb{R}^n$ and $\alpha_1, \alpha_2, \alpha_3 \in \mathcal{K}_\infty$, then $x_+ = f(x)$ is GAS.

^aThis result also holds for the time-varying system $x_+ = f_t(x_t)$.

Proof. We give the proof under a simplifying assumption, although the result holds in general. Suppose that there exists $c \in (0, 1]$ such that $\alpha_3(r) \geq c\alpha_2(r)$ for all r . Then

$$cV(x) \leq c\alpha_2(\|x\|) \leq \alpha_3(\|x\|), \quad (3.14)$$

so that

$$V(f(x)) \leq V(x) - \alpha_3(\|x\|) \leq (1 - c)V(x). \quad (3.15)$$

Iterating, $V_t \leq (1 - c)^t V_0$, and hence

$$\alpha_1(\|x_t\|) \leq (1 - c)^t \alpha_2(\|x_0\|) \implies \|x_t\| \leq \alpha_1^{-1}\left((1 - c)^t \alpha_2(\|x_0\|)\right) =: \beta(\|x_0\|, t). \quad (3.16)$$

By the properties of Lemma 3.13, $\beta(r, s)$ is a class $\mathcal{K}$ (or $\mathcal{K}_\infty$) function of r for any fixed s , and $\beta(r, s) \rightarrow 0$ as $s \rightarrow \infty$ for any fixed r , and thus is class $\mathcal{KL}$. $\square$

Lemma 3.11 (Polynomial bounds imply GES). *Suppose $\alpha_i(\|x\|) = \bar{\alpha}_i \|x\|^p$ for $i = 1, 2, 3$, with $\bar{\alpha}_1 \leq \bar{\alpha}_2$ and $\bar{\alpha}_3 \leq \bar{\alpha}_2$, and suppose*

$$\bar{\alpha}_1 \|x\|^p \leq V(x) \leq \bar{\alpha}_2 \|x\|^p, \quad V(f(x)) - V(x) \leq -\bar{\alpha}_3 \|x\|^p. \quad (3.17)$$

Then $x_+ = f(x)$ is GES.

Proof. From the hypotheses,

$$V_+ \leq V - \bar{\alpha}_3 \|x\|^p \leq \left(1 - \frac{\bar{\alpha}_3}{\bar{\alpha}_2}\right) V, \quad (3.18)$$

so that $V_t \leq (1 - \bar{\alpha}_3/\bar{\alpha}_2)^t V_0$, and hence

$$\|x_t\| \leq \left(\frac{\bar{\alpha}_2}{\bar{\alpha}_1}\right)^{1/p} \left(1 - \frac{\bar{\alpha}_3}{\bar{\alpha}_2}\right)^{t/p} \|x_0\|. \quad (3.19)$$

□

Example 3.12 (Linear systems). Consider the linear system $x_+ = Ax$ with $V(x) = x^\top Px$, where P satisfies the discrete-time Lyapunov equation

$$A^\top PA - P + Q = 0, \quad Q \succ 0. \quad (3.20)$$

Such a solution P always exists whenever A is (Schur) stable, and is given by the observability gramian

$$P = \sum_{k=0}^{\infty} (A^\top)^k Q A^k. \quad (3.21)$$

It follows that $\lambda_{\min}(P) \|x\|^2 \leq V(x) = x^\top Px \leq \lambda_{\max}(P) \|x\|^2$, and

$$V(Ax) - V(x) = x^\top (A^\top PA - P)x = -x^\top Qx \leq -\lambda_{\min}(Q) \|x\|^2. \quad (3.22)$$

The lemma above then applies with $p = 2$, $\bar{\alpha}_1 = \lambda_{\min}(P)$, $\bar{\alpha}_2 = \lambda_{\max}(P)$, and $\bar{\alpha}_3 = \lambda_{\min}(Q)$ (one checks $\lambda_{\min}(Q) \leq \lambda_{\max}(P)$), recovering exponential stability from the spectral condition in Key Result 3.1.

3.6 Technical Lemmas

Lemma 3.13 (Properties of class $\mathcal{K}$, $\mathcal{K}_\infty$, and $\mathcal{KL}$ functions, cf. Khalil [Kha02, Lemma 4.2]). *Let α_1 and α_2 be class $\mathcal{K}$ functions on $[0, a)$, α_3 and α_4 be class $\mathcal{K}_\infty$ functions, and β be a class $\mathcal{KL}$ function. Denote the inverse of α_i by α_i^{-1} . Then,*

- α_1^{-1} is defined on $[0, \alpha_1(a))$ and belongs to class $\mathcal{K}$.
- α_3^{-1} is defined on $[0, \infty)$ and belongs to class $\mathcal{K}_\infty$.
- $\alpha_1 \circ \alpha_2$ belongs to class $\mathcal{K}$.
- $\alpha_3 \circ \alpha_4$ belongs to class $\mathcal{K}_\infty$.
- $\sigma(r, s) = \alpha_1(\beta(\alpha_2(r), s))$ belongs to class $\mathcal{KL}$.

3.7 Further Reading

The classical reference for the material in this chapter – and for essentially everything Lyapunov-theoretic we do for the rest of the course – is Khalil [Kha02]; note, however, that Khalil works in continuous time throughout, so every definition and result above is our own discrete-time translation of his. For a treatment that stays in discrete time from the outset, Bof, Carli, and Schenato [BCS18] collect and prove exactly this class of results – comparison-function characterizations of stability, exponential stability, the direct method, and more – including discrete-time versions of several of the technical lemmas we used but did not prove above. Boyd [Boy08a] is a compact, example-driven pass over the same continuous-time material as Khalil’s Chapter 4 (the direct method, invariance, exponential stability, converse theorems), and is a good place to build intuition before

returning to the more careful treatment here; its companion lecture, Boyd [Boy08b], works out the linear/quadratic special case in detail, including the (continuous- and discrete-time) Lyapunov equation and its gramian solution, directly paralleling the worked linear-system example above.

References

- [BCS18] Nicoletta Bof, Ruggero Carli, and Luca Schenato. *Lyapunov Theory for Discrete Time Systems*. arXiv:1809.05289. Sept. 2018. URL: <https://arxiv.org/abs/1809.05289> (visited on 08/20/2026).
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- [Kha02] Hassan K. Khalil. *Nonlinear Systems*. 3rd. Upper Saddle River, NJ: Prentice Hall, 2002.